Input system:
Dependent variables:x_1, x_2
Independent variables:t
Equations:

$$\frac{dx_1}{dt} = ax_1^2 \mathrm{inv}\left(N + x_1\right) - bx_1x_2$$

$$\frac{dx_2}{dt} = \left(-1\right) cx_2 + dx_1x_2$$

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Result:
Dependent variables:s_1, x_1, x_2
Independent variables:t
Additional variables:

$$s_1 = \mathrm{inv}\left(N + x_1\right)$$

Equations:

$$\frac{dx_1}{dt} = ax_1^2 s_1 + \left(-1\right) bx_1x_2$$

$$\frac{dx_2}{dt} = dx_1x_2 + \left(-1\right) cx_2$$

$$\frac{ds_1}{dt} = \left(-1\right) s_1^3 ax_1^2 + s_1^2 bx_1x_2$$